

GMaster: a difference-form Wigner- d march for GPU MASTER estimation

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ABSTRACT

The latitudinal factor of a HEALPix spherical-harmonic transform is a three-term Jacobi recurrence in ℓ for each order m . In float32 that recurrence is unstable near the poles and at every turning point: the characteristic roots collide, and a rounding of size $u|v|$ is amplified by $1/\sin\phi$. We rewrite the march so that the carried pair is (v_n, D_n) with $D_n = v_n - v_{n-1}$ rather than (v_n, v_{n-1}) . The difference coefficient vanishes as ϕ^2 exactly where the plain form is ill-conditioned, error is injected at the scale of $D \sim \phi v$, and the float32 floor becomes the random walk $\sqrt{n}u$. Against the float64 recurrence at $N_{\text{side}} = 1024$ the maximum relative error over every lane falls from 4.2×10^{-2} (plain) to 2.8×10^{-6} (shipped). The identities of the rewrite, a southern-hemisphere reflection onto the same code path, and an exact per-block power-of-two rescale are proved in Lean 4. The kernel is CUDA, called from JAX through the XLA FFI, and is the default latitudinal operator in GMaster, a GPU library with the public API of NaMaster. On one RTX PRO 6000 Blackwell against NaMaster on a 96-core Threadripper PRO 9995WX the end-to-end MASTER pipeline is $10\text{--}21\times$ faster at $N_{\text{side}} = 128\text{--}2048$ in both spins, with coupled spectra agreeing to $1.2 \times 10^{-7}\text{--}1.2 \times 10^{-5}$. On the ACT DR6 night PA4 $f150$ coadd and a FLAMINGO L2p8 Compton- y map, both at $N_{\text{side}} = 4096$, warm wall-clock is $7.6\text{--}8.6\text{ s}$ against 70 s for NaMaster, with fractional bandpower rms 1.4×10^{-5} and 2.3×10^{-7} . No other GPU spherical-harmonic code was timed.

Key words: methods: numerical – cosmology: cosmic microwave background – cosmology: large-scale structure of Universe

1 INTRODUCTION

The MASTER estimator (Hivon et al. 2002) recovers binned angular power spectra from masked spherical maps by analysing the masked fields, forming coupled pseudo-spectra, building a mode-coupling matrix from the mask, and decoupling. NaMaster (Alonso et al. 2019) is the CPU reference. Its transforms run through DUCC, the successor of libsharp (Reinecke & Seljebotn 2013), on the HEALPix pixelisation (Górski et al. 2005). The coupling algebra has a structured Wigner-3j recurrence (Kiddier & Gratton 2026).

On a GPU the same two cubic stages remain. An on-the-fly associated-Legendre recurrence, as in SHTns (Schaeffer 2013) and DUCC, is a serial chain in ℓ with a reduction over colatitude at every degree. That pattern matches AVX-512 CPUs and mismatches GPU SIMT. A precomputed Legendre matrix, as in the precompute mode of s2fft (Price & McEwen 2024), inverts the bottleneck provided the table fits. Related GPU work (Belkner et al. 2024; Reinecke et al. 2023) retains an on-the-fly latitudinal factor and is not a drop-in for a full-sky HEALPix MASTER pipeline.

GMaster is a JAX implementation of the NaMaster public API. This Letter isolates one change that made a table-free

float32 latitudinal kernel accurate enough to be the default at every $N_{\text{side}} \geq 64$: marching the difference pair of the Jacobi row attached to $d_{m,-s}^\ell(\theta)$ rather than the consecutive values. The recurrence, a reflection that sends the southern hemisphere through the same instructions, and the exactness of a per-block power-of-two rescale are proved in Lean 4 (de Moura & Ulrich 2021; The mathlib Community 2020). We report the end-to-end MASTER board and two real-map overlays. This is not a cosmological analysis: no beam, split cross-spectrum, or covariance is applied to the survey maps.

2 DIFFERENCE-FORM MARCH

Every Wigner function $d_{m,-s}^\ell(\theta)$ is an algebraic multiple of a Jacobi polynomial (Olver et al. 2010, NIST DLMF 14.9.16, 18.3),

$$d_{m,-s}^\ell(\theta) = (-1)^m 2^{\varepsilon_m \text{LG2N}} \times \left(\sin \frac{\theta}{2}\right)^\alpha \left(\cos \frac{\theta}{2}\right)^\beta P_n^{(\alpha,\beta)}(\cos \theta), \quad (1)$$

with $\alpha = m + s$, $\beta = |m - s|$, $n = \ell - \max(m, s)$ and $\text{LG2N} = \log_2 \sqrt{(\ell + m)! (\ell - m)! / ((\ell - s)! (\ell + s)!)}$. The family obeys a three-term recurrence in n (Olver et al. 2010, DLMF 18.9.1).

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Writing $x = \cos \theta$ and v_n for the marched polynomial,

$$v_n = A_n v_{n-1} - B_n v_{n-2}, \quad A_n = c_1 x + c_0, \quad B_n = c_2, \quad (2)$$

where c_1, c_0, c_2 depend only on (ℓ, m) . A float64 march of (2) agrees with a numpy oracle of the same coefficients to 1.5×10^{-15} at $N_{\text{side}} = 1024$. Plain float32 of the same recurrence reaches a maximum relative error of 4.2×10^{-2} over the row (Table 1).

Over a few degrees the coefficients are nearly constant and $c_2 \rightarrow 1$, so the row is locally $v_n \approx A v_{n-1} - v_{n-2}$ with characteristic roots $e^{\pm i\phi}$ and $2\sqrt{c_2} \cos \phi = A$. In the oscillatory region $|A| < 2$, and $\phi \rightarrow 0$ both at the poles and at every turning point $m \approx (\ell + \frac{1}{2}) \sin \theta$. A rounding $\delta \sim 2u|v|$ of $\text{fl}(A v_{k-1} - B v_{k-2})$ perturbs the state by $(\delta, 0)$, which decomposes on the modes $(e^{\pm i\phi}, 1)$ with amplitude $\delta/(2 \sin \phi)$ each and evolves as $\delta \sin((n-k+1)\phi)/\sin \phi$: every rounding is amplified by $1/\sin \phi$, and over n steps with random signs the relative error is $\sim \sqrt{n} u / \sin \phi$. On the first HEALPix ring at $N_{\text{side}} = 4096$, $\theta \approx 2 \times 10^{-4}$ and $1/\sin \phi \approx 5 \times 10^3$, giving a floor of a few per cent, the order of the plain-float32 row of Table 1. A systematic error is worse: a relative error ϵ in x perturbs A at every step, and $2 \cos \phi = A$ turns it into a coherent phase drift $\sim n c_1 x \epsilon / (2 \sin \phi)$, of order a radian at $n = 12288$ for $\epsilon = u$.

Let $D_n = v_n - v_{n-1}$. Then for $n \geq 2$

$$D_n = (A_n - 1 - B_n) v_{n-1} + B_n D_{n-1}, \quad v_n = v_{n-1} + D_n. \quad (3)$$

In terms of ϕ , the difference coefficient is $C = A - 1 - c_2 \approx -\phi^2$, so $D \approx \phi |v|$ exactly where the plain march is ill-conditioned. A rounding of $v \leftarrow \text{fl}(v + D)$ lands on v with D fixed and is not amplified. A rounding of $D \leftarrow \text{fl}(c_2 D + C v)$ lands on D at scale $u\phi|v|$ and, after the $1/(2 \sin \phi)$ modal factor, contributes $O(u|v|)$ per step. Both kinds therefore give a random-walk floor $\sqrt{n} u$ independent of ϕ : $\sqrt{3072} u \approx 3.3 \times 10^{-6}$ at $N_{\text{side}} = 1024$, against the measured 2.8×10^{-6} .

The only coefficient on the amplified path is C , and the argument needs its error to be $O(u|C|)$, not $O(u|A|)$. Writing $C_n = c_1(x-1) + E_n$ with $E_n = c_1 - 1 - c_2 + c_0$, the kernel carries $x-1$ rather than x : near a pole $x-1 \sim -\theta^2/2$, so a single-word x would inject a relative error $u/(\theta^2/2)$ into the leading term of C , and a single-word E would leave an error $\sim u|c_1|$ that does not shrink with C . Both are systematic and drift the phase. The kernel therefore stores $|x| - 1 = x_h + x_l$, $c_1 = c_{1h} + c_{1l}$ and $E = E_h + E_l$ as float32 pairs, each split from float64 to 2^{-48} (Dekker 1971), and forms

$$C = \text{fma}(c_{1h}, x_h, E_h) + [\text{fma}(c_{1h}, x_l, \text{fma}(c_{1l}, x_h, E_l))]. \quad (4)$$

The first **fma** rounds once, by $u|C|$, and that rounding vanishes with C at the turning point where the amplification peaks; the bracket is $\sim 2^{-24}$ of the leading terms. A single-word C leaves the coherent $u|c_1|$ -class drift, growing like n rather than $\sqrt{n}$: the 4×10^{-5} floor in the middle row of Table 1. The coefficient c_2 is single-word, because its error enters as $|D| \sim u\phi|v|$ on D , the unamplified kind.

The southern hemisphere needs no second code path. If $v_n = (c_1 x + c_0) v_{n-1} - c_2 v_{n-2}$ then $w_n := (-1)^n v_n$ satisfies

$$w_n = (c_1(-x) - c_0) w_{n-1} - c_2 w_{n-2}, \quad (5)$$

the same recurrence with $y = |x|$ for x and $-c_0$ for c_0 , whose difference coefficient is $c_1(y-1) + E_n^-$ with $E_n^- = c_1 - 1 - c_2 - c_0$. Every lane, northern or southern, therefore marches

Table 1. Maximum relative error of a float32 march against the float64 recurrence, over every lane, at $N_{\text{side}} = 1024$ spin 0. The two-word row is the shipped kernel (5.7×10^{-6} at $N_{\text{side}} = 4096$).

Float32 march	$m = 0$	$m = 1000$	$m = L/2$
Plain, single word	4.2×10^{-2}	8.8×10^{-5}	3.1×10^{-5}
Difference form, 1-word C	4.1×10^{-5}	8.6×10^{-5}	5.1×10^{-5}
Difference form, 2-word C	2.8×10^{-6}	2.9×10^{-6}	1.6×10^{-5}

$|x| - 1 \in (-1, 0]$ with the same instructions and one of two table columns $E^\pm$, and the emit restores the sign on odd n . Each lane stores $v \cdot 2^{ex}$ with ex an integer and is rescaled by an exact power of two every eight degrees; the map $(v, D) \mapsto (v + D', D')$ is linear, so the represented value is unchanged, and the per-degree growth is bounded by $1 + |C| + c_2 \leq 2^{15}$, so eight degrees cannot overflow. The emit factorises the Jacobi normalisation $2^{\epsilon_m \text{LG}2N}$ into a per-lane power of two and a per-degree mantissa proved to lie in $[2^{-45.5}, 2)$, so no exponent lane leaves the kernel. Equations (3) and (5), the seeds, the homogeneity of the rescale, and the range of the mantissa are theorems in a Lean 4 development on Mathlib that builds with no **sorry**; the modal error argument is not formalised, and is checked by Table 1.

The kernel is CUDA, one warp per $(m, \text{ring tile})$, exposed as XLA FFI handlers inside **jit**. A field and its mask are two transforms of the same rows: the analysis kernel takes both as emit channels, so an **NmtField** is not two launches. For $s = 0$ the southern hemisphere is folded into the northern recurrence. The default at $L = \ell_{\text{max}} + 1 \geq 192$ ($N_{\text{side}} \geq 64$) is this march; an exact float64 route remains available by flag. The difference-form D_n march is not GPU-intrinsic; only the shipped latitudinal operator is CUDA. A fair CPU timing of the same algorithm requires porting that kernel, not the JAX CPU fallback with the CUDA path gated out.

3 PERFORMANCE

All timings are single-tenant on one NVIDIA RTX PRO 6000 Blackwell. The CPU reference is NaMaster 3.0 / **ducc0** 0.39.1 on the same host, a 96-core Threadripper PRO 9995WX. End-to-end totals are warmed medians of field construction, coupling matrix, coupled cell and decoupling. Relative error is the maximum relative deviation of the coupled spectrum from **pymaster**. Isolated transforms are timed against the **ducc0** call NaMaster issues.

Table 2 is the MASTER board. The shipped route is 10.2–21.2× NaMaster at $N_{\text{side}} = 128$ –2048 in both spins, 5.0–5.8× at $N_{\text{side}} = 64$ (launch-latency bound) and 9.6× at $N_{\text{side}} = 4096$ spin 0. Stock NaMaster segfaults at $N_{\text{side}} = 4096$ spin 2 on 32-bit coupling-matrix indices; GMaster finishes that cell in 8.8 s. One isolated latitudinal pass against **ducc0** is 5.6–8.5× at $N_{\text{side}} = 1024$ –4096, with coefficient agreement $\sim 1.2 \times 10^{-7}$. With the march disabled and rings and coupling forced to float64, decoupled bandpowers reproduce 4.8×10^{-13} at $N_{\text{side}} = 256$ spin 0. The shipped default is a float32-class route: $\sim 10^{-6}$ relative on the decoupled C_ℓ , five to six orders below NaMaster's own $n_{\text{iter}} = 3$ truncation.

Figure 1 is the same estimator on the ACT DR6 night PA4 f150 source-free coadd (Naess et al. 2025), Stokes I , degraded from native $N_{\text{side}} = 8192$ to 4096. The mask is the survey window (finite non-zero pixels; $f_{\text{sky}} = 0.481$). GMaster

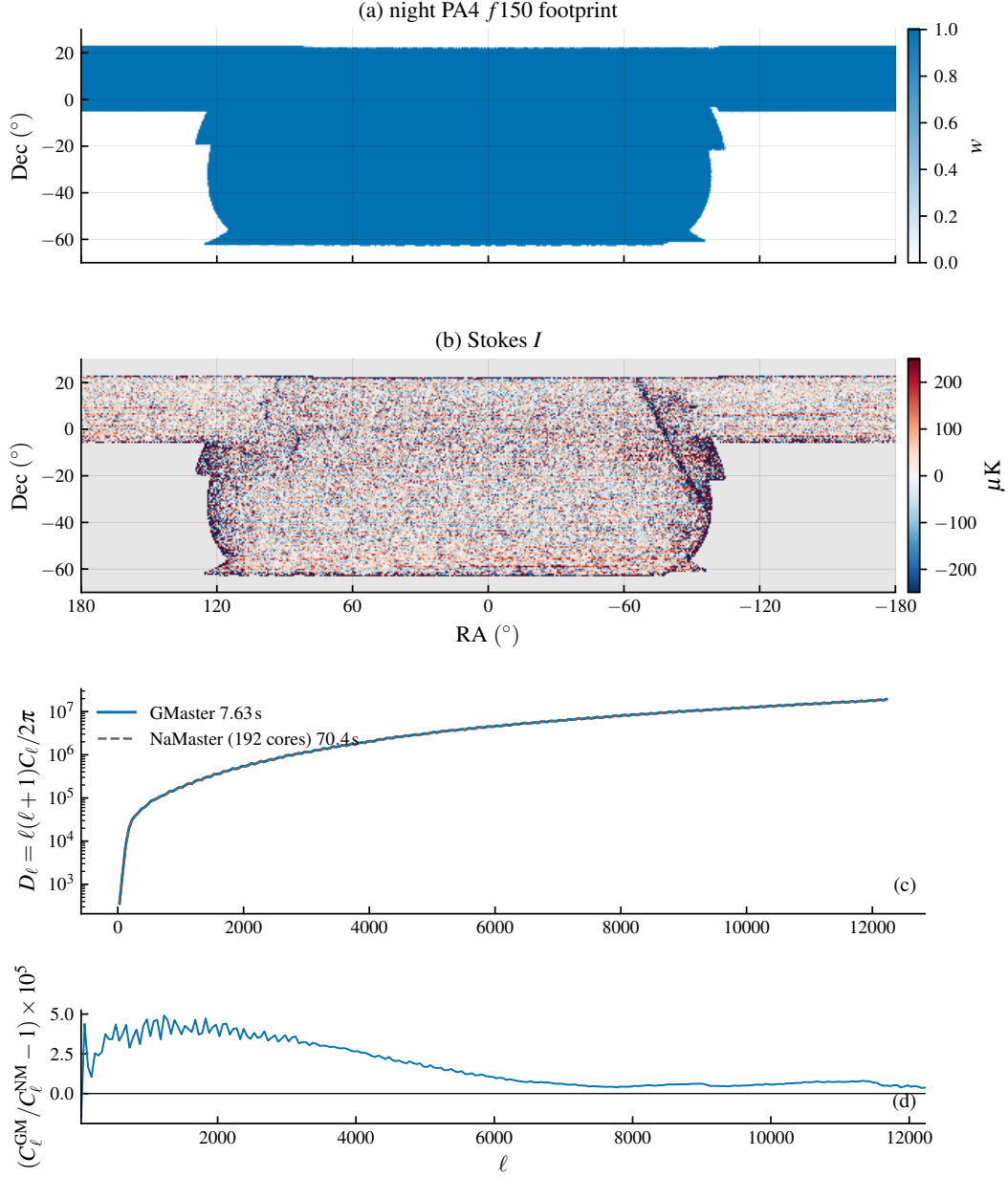


Figure 1. ACT DR6 night PA4 $f150$ source-free coadd (Naess et al. 2025) at $N_{\text{side}} = 4096$. (a) Survey window used as the mask ($f_{\text{sky}} = 0.481$). (b) Stokes I on that footprint. (c) Decoupled TT D_ℓ from GMaster and NaMaster, linear bins of width 50, $n_{\text{iter}} = 3$. (d) Fractional difference. This is a methods overlay, not an ACT science product.

and NaMaster share the mask, linear bins of width 50, and $n_{\text{iter}} = 3$. Decoupled TT agrees to $\text{rms}(C_\ell^{\text{GM}}/C_\ell^{\text{NM}} - 1) = 1.41 \times 10^{-5}$. Wall-clock after compilation: 7.63 s versus 70.4 s on 192 cores. The FLAMINGO L2p8 unlensed Compton- y lightcone (Schaye et al. 2023), full sky at $N_{\text{side}} = 4096$, gives rms 2.32×10^{-7} in 8.64 s versus 69.5 s. Both overlays time only the estimator. Figure 2 puts the float32 floors of Table 1 beside pipeline-level agreement of the shipped and exact routes, and Figure 3 is the wall-clock board of Table 2.

4 CONCLUSIONS

A three-term Jacobi recurrence in float32 is limited by $1/\sin \phi$ amplification, not by the width of its words. Marching $(v, v - v_{\text{prev}})$ moves the injection of every rounding onto a difference that already vanishes at the turning points, and a two-word C removes the residual coherent drift. Carried as a CUDA kernel behind NaMaster's API, that rewrite is the default latitudinal operator in GMaster and is $10\text{--}21\times$ NaMaster at $N_{\text{side}} = 128\text{--}2048$ on the board of Table 2. The same pipeline on an ACT DR6 coadd and a FLAMINGO Compton- y map at $N_{\text{side}} = 4096$ agrees at $10^{-5}\text{--}10^{-7}$ in under nine seconds.

No other GPU spherical-harmonic implementation was timed. The shipped default trades NaMaster's float64 con-

Table 2. End-to-end MASTER on the 96-core host. NM and GM are NaMaster and GMaster milliseconds. rel is the maximum relative deviation of the coupled spectrum from `pymaster`.

N_{side}	spin	NM	GM	Ratio	rel
64	0	9	2	5.8	1.15×10^{-7}
64	2	12	2	5.0	2.72×10^{-7}
128	0	20	1	13.3	4.92×10^{-7}
128	2	40	3	13.1	4.37×10^{-7}
256	0	58	4	15.0	6.36×10^{-7}
256	2	161	8	20.7	6.30×10^{-7}
512	0	314	17	18.1	7.07×10^{-7}
512	2	641	30	21.2	2.99×10^{-6}
1024	0	1682	102	16.5	1.45×10^{-6}
1024	2	3017	212	14.2	7.79×10^{-6}
2048	0	10311	1014	10.2	7.25×10^{-6}
2048	2	17767	1244	14.3	1.19×10^{-5}
4096	0	71583	7458	9.6	7.0×10^{-6}
4096	2	—	8805	—	finite

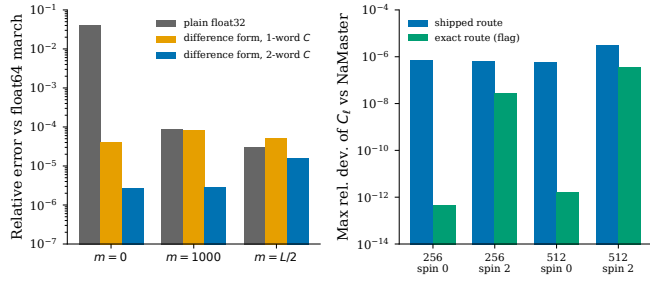


Figure 2. Left: relative error of a float32 march against the float64 recurrence at $N_{\text{side}} = 1024$ spin 0. Right: maximum relative deviation of the decoupled bandpowers from NaMaster for the shipped route and the exact float64 route.

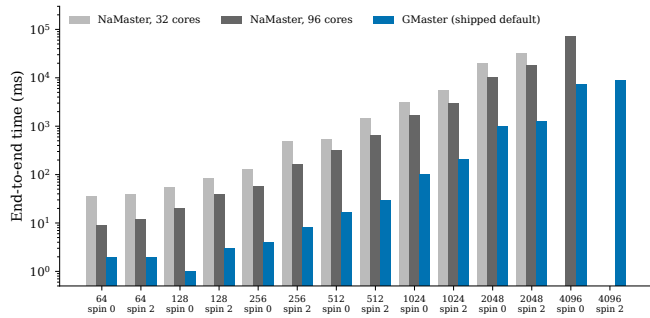


Figure 3. End-to-end MASTER wall time on the host of Table 2. Grey is NaMaster on 96 cores; blue is GMaster. NaMaster is absent at $N_{\text{side}} = 4096$ spin 2, where stock NaMaster segfaults on 32-bit coupling-matrix indices.

tract for $\sim 10^{-6}$ relative agreement; the exact route is one flag. The overlays are methods comparisons, not science products.

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NaMaster, DUCC/libsharp, s2fft, threej_cosmo and SHTns are the reference stack against which the numbers were mea-

sured. The Lean 4 development uses Mathlib. ACT DR6 maps are those of Naess et al. (2025). Portions of the library and of this manuscript were drafted with coding assistants; every numerical claim was measured on the hardware of Section 3.

DATA AVAILABILITY

GMaster, the overlay scripts, the committed bandpowers of both overlays and a notebook that rebuilds Figure 1 and the timing table from them are at <https://github.com/licongxu/GMaster>. Figure 1 uses the public ACT DR6 maps (Naess et al. 2025). The FLAMINGO L2p8 Compton- y light-cone is described by Schaye et al. (2023).

REFERENCES

- Alonso D., Sánchez J., Slosar A., 2019, *Monthly Notices of the Royal Astronomical Society*, 484, 4127
- Belkner S., Duivenvoorden A. J., Carron J., Schaeffer N., Reinecke M., 2024, *arXiv e-prints*
- Dekker T. J., 1971, *Numerische Mathematik*, 18, 224
- Górski K. M., Hivon E., Banday A. J., Wandelt B. D., Hansen F. K., Reinecke M., Bartelmann M., 2005, *The Astrophysical Journal*, 622, 759
- Hivon E., Górski K. M., Netterfield C. B., Crill B. P., Prunet S., Hansen F., 2002, *The Astrophysical Journal*, 567, 2
- Kiddier G., Gratton S., 2026, *Monthly Notices of the Royal Astronomical Society*, 549, stag1075
- Naess S., Guan Y., Duivenvoorden A. J., Hasselfield M., Wang Y., et al., 2025, *Journal of Cosmology and Astroparticle Physics*, 2025, 061
- Olver F. W. J., Lozier D. W., Boisvert R. F., Clark C. W., eds, 2010, *NIST Handbook of Mathematical Functions*. Cambridge University Press
- Price M. A., McEwen J. D., 2024, *Journal of Computational Physics*, 510, 113109
- Reinecke M., Seljebotn D. S., 2013, *Astronomy & Astrophysics*, 554, A112
- Reinecke M., Belkner S., Carron J., 2023, *Astronomy & Astrophysics*, 678, A165
- Schaeffer N., 2013, *Geochemistry, Geophysics, Geosystems*, 14, 751
- Schaye J., Kugel R., Schaller M., Helly J. C., van Loon J., et al., 2023, *Monthly Notices of the Royal Astronomical Society*, 526, 4978
- The mathlib Community 2020, in *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*. ACM, pp 367–381, doi:10.1145/3372885.3373824
- de Moura L., Ullrich S., 2021, in *Automated Deduction – CADE 28*. Springer, pp 625–635, doi:10.1007/978-3-030-79876-5_37

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